

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: *Measure network performance using key metrics with simulation tools*

Assessment Tool Weightage: 20%

QUESTIONS: 40

Total Marks:40

Annexure A

Mock Interview

QUESTIONS:40

Total Marks:40

1. Analyze how network congestion affects packet delay and packet loss. (BL4)(1)
2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)(1)
3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)(1)
4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)(1)
5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)(1)
6. Analyze how network segmentation using VLANs improves performance and security. (BL4)(1)
7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)(1)
8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)(1)
9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)(1)
10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)(1)
11. A network has high delay—how will you identify and fix the issue? (BL4)(1)
12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)(1)
13. How would you reduce packet loss in a congested network? (BL4)(1)
14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)(1)
15. Given a network with high jitter, suggest improvements (BL4)(1)
16. How will you simulate a congested network in Packet Tracer? (BL5)(1)
17. If packets are delayed in a router, what steps will you take? (BL4)(1)
18. How would you optimize a network for online gaming? (BL4)(1)
19. Design a small network and explain how you will measure its performance (BL5)(1)
20. If RTT is very high, how will you diagnose the issue? (BL4)(1)
21. Explain network performance to a non-technical person (BL5)(1)
22. Describe throughput using a real-life example (BL5)(1)
23. Present how packet loss affects video streaming (BL5)(1)
24. Explain delay types clearly in under 1 minute (BL4)(1)
25. Describe Cisco Packet Tracer in a professional way (BL5)(1)
26. Explain QoS in simple and technical terms (BL5)(1)

- 27. How would you present your simulation results to a client? (BL4)(1)
- 28. Explain jitter using a real-world analogy (BL5)(1)
- 29. Describe how you tested network performance in a lab (BL4)(1)
- 30. Explain the importance of performance metrics in one structured answer (BL5)(1)
- 31. Introduce yourself and explain your knowledge of network performance (BL5)(1)
- 32. Why are you interested in networking and simulation tools? (BL5)(1)
- 33. What makes you confident in analyzing network performance? (BL5)(1)
- 34. Explain a concept you are very strong in (related to networking) (BL5)(1)
- 35. How would you handle a question you don't know? (BL5)(1)
- 36. Describe a situation where you solved a technical problem (BL4)(1)
- 37. Why should we select you for a networking role? (BL5)(1)
- 38. What challenges did you face while learning Packet Tracer? (BL5)(1)
- 39. How do you improve your technical skills regularly? (BL5)(1)
- 40. Demonstrate confidence while explaining a complex concept simply (BL5)(1)

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

Mock Interview Answers

1. How does network congestion affect packet delay and packet loss?

Network congestion occurs when traffic exceeds network capacity. It increases queueing delay in routers and may cause buffer overflow, resulting in packet loss.

2. Compare star and mesh topologies on reliability and fault tolerance.

Star topology is easier to manage but depends on the central switch. Mesh topology provides multiple paths, so failure of one link does not usually interrupt communication. Therefore, mesh has higher fault tolerance.

3. Why does increasing routers between source and destination increase end-to-end delay?

Each router introduces processing, queuing, transmission and propagation delays. Therefore, adding more routers generally increases the total end-to-end delay.

4. How does QoS improve voice and video traffic?

QoS prioritizes delay-sensitive traffic such as voice and video. It reduces delay, jitter and packet loss, resulting in smoother calls and video streaming.

5. Evaluate fiber optic instead of copper for high-speed communication.

Fiber provides higher bandwidth, longer transmission distance, lower signal loss and immunity to electromagnetic interference. Its disadvantages are higher installation cost and more complex maintenance.

6. How does VLAN segmentation improve performance and security?

VLANs divide a network into logical segments. This reduces unnecessary broadcast traffic and restricts communication between departments, improving both performance and security.

7. Compare TCP and UDP.

TCP is connection-oriented and provides reliable, ordered delivery with error recovery. UDP is connectionless and has lower overhead, making it suitable for real-time applications such as VoIP, gaming and streaming.

8. Why is load balancing important in enterprise networks?

Load balancing distributes traffic across multiple links or servers. It prevents individual resources from becoming overloaded and improves performance, availability and reliability.

9. How do frequent routing updates influence network efficiency?

Routing updates consume bandwidth and router processing resources. Excessive updates can reduce efficiency, although timely updates are necessary to maintain accurate routing information.

10. What is the trade-off between security and network performance?

Firewalls, encryption and inspection improve security but require processing resources and can introduce some latency. Proper configuration balances security requirements with acceptable performance.

11. A network has high delay. How will you identify and fix it?

I would use **ping, traceroute and Packet Tracer simulation** to identify where delay occurs. I would then check congestion, routing paths, bandwidth and overloaded devices and optimize the affected section.

12. If throughput is low despite high bandwidth, what could be the reasons?

Possible reasons include packet loss, congestion, high latency, protocol overhead, hardware limitations, poor configuration or an inefficient routing path.

13. How would you reduce packet loss in a congested network?

I would reduce congestion using QoS, traffic prioritization, load balancing, additional bandwidth and better routing. I would also check faulty cables and overloaded network devices.

14. A VoIP call is breaking frequently. How would you troubleshoot?

I would check packet loss, jitter, latency and bandwidth using network monitoring tools. Then I would verify QoS configuration, network congestion, routing and device performance.

15. Given a network with high jitter, suggest improvements.

I would apply QoS, prioritize real-time traffic, reduce congestion, optimize routing and use sufficient bandwidth. Load balancing can also help distribute traffic.

16. How will you simulate a congested network in Packet Tracer?

I would create multiple devices and generate traffic through a limited-bandwidth link. I would monitor packet transmission, delay, packet loss and throughput to observe the effect of congestion.

17. If packets are delayed in a router, what steps will you take?

I would check router CPU utilization, interface utilization, queue length, routing configuration and bandwidth. I would use ping and traceroute to locate the delay and then optimize routing or traffic management.

18. How would you optimize a network for online gaming?

I would minimize latency and jitter, use QoS to prioritize gaming traffic, avoid congested routes, use reliable high-speed connectivity and ensure the router is not overloaded.

19. Design a small network and explain how you will measure performance.

I would create a **star topology** using a switch, router and end devices. I would measure **throughput, delay, packet loss and jitter** using ping, simulation tools and traffic analysis.

20. If RTT is very high, how will you diagnose the issue?

I would use ping and traceroute to identify the destination and hop causing high RTT. I would then check congestion, routing, bandwidth, link quality and device processing delays.

21. Explain network performance to a non-technical person.

Network performance means **how quickly and reliably information travels through a network**. A good network delivers data quickly, with little waiting, interruption or loss.

22. Describe throughput using a real-life example.

Throughput is the actual amount of data successfully transferred per unit time. For example, if a network successfully transfers 100 MB in 10 seconds, its throughput is approximately 10 MB/s.

23. How does packet loss affect video streaming?

Packet loss causes missing data during transmission. This can produce buffering, freezing, reduced video quality and interruptions in streaming.

24. Explain delay types in under one minute.

The main delays are **processing, queuing, transmission and propagation delay**. Processing occurs when devices handle packets, queuing occurs while packets wait, transmission is the time to place bits on the link, and propagation is the time for the signal to travel through the medium.

25. Describe Cisco Packet Tracer professionally.

Cisco Packet Tracer is a network simulation tool used to design, configure and test network topologies virtually. It allows users to practice device configuration and analyze network behavior without requiring physical hardware.

26. Explain QoS in simple and technical terms.

Simply, QoS means **giving important traffic priority**. Technically, QoS uses classification, prioritization and traffic-management mechanisms to control delay, jitter, packet loss and bandwidth usage.

27. How would you present simulation results to a client?

I would present the topology, objectives, measured metrics, results and key observations using tables or graphs. I would compare the results with expected performance and provide practical recommendations.

28. Explain jitter using a real-world analogy.

Jitter is variation in packet arrival time. Imagine a bus that should arrive every 10 minutes but sometimes arrives after 5 minutes and sometimes after 20 minutes. That irregular timing is similar to network jitter.

29. Describe how you tested network performance in a lab.

I would create the topology in Packet Tracer, configure IP addresses and routing, generate traffic and measure throughput, delay, packet loss and jitter. I would compare results under normal and congested conditions.

30. Explain the importance of performance metrics.

Performance metrics such as **throughput, delay, packet loss and jitter** help determine whether a network is operating efficiently. They also help identify problems and guide network optimization.

31. Introduce yourself and explain your knowledge of network performance.

Answer:

“I am a computer engineering student with an interest in networking. I understand important performance metrics such as throughput, delay, packet loss and jitter. I have also learned to design and analyze networks using simulation tools such as Cisco Packet Tracer.”

32. Why are you interested in networking and simulation tools?

“I am interested in networking because it explains how devices communicate and share information. Simulation tools are useful because they allow me to design, configure and troubleshoot networks safely before implementing them physically.”

33. What makes you confident in analyzing network performance?

“My confidence comes from understanding performance metrics and using a systematic troubleshooting approach. I analyze throughput, delay, packet loss and jitter to identify the cause of performance problems.”

34. Explain a concept you are very strong in.

“One concept I am comfortable with is **QoS**. QoS classifies and prioritizes important traffic, especially voice and video, so that delay, jitter and packet loss can be controlled.”

35. How would you handle a question you don't know?

“I would be honest that I do not know the answer completely. I would explain what I understand about the related concept and then verify the correct information rather than providing an incorrect answer.”

36. Describe a situation where you solved a technical problem.

“In a network simulation, if devices could not communicate, I would systematically check IP addresses, subnet masks, interfaces, cables and routing configuration. By testing each layer step by step, I can identify and correct the problem.”

37. Why should we select you for a networking role?

“I have a good understanding of networking concepts and performance metrics, and I approach problems logically. I am also willing to learn new technologies and improve my practical networking skills.”

38. What challenges did you face while learning Packet Tracer?

“Initially, configuring IP addresses, routing and understanding device connections was challenging. I overcame this by practicing different topologies and troubleshooting configuration errors systematically.”

39. How do you improve your technical skills regularly?

“I improve my skills through regular practice, simulations, technical documentation, problem-solving exercises and small networking projects. I also review mistakes and learn from them.”

40. Demonstrate confidence while explaining a complex concept simply.

“Network congestion is similar to traffic on a busy road. When too many vehicles use the same road, movement becomes slower. Similarly, when too much data uses a network, packets wait longer, causing delay and sometimes packet loss.”